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Software Requirement Specification (SRS)

PSB Hackathon 2026

Software Requirement Specification
Version 1

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Team Name: ASE
Institute Name: Punjab Engineering College

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Declaration

The purpose of this Software Requirements Specification (SRS) document is to identify and document the user requirements for the **"Quantum-Proof Systems Scanner"** The end deliverable software that will be supplied by **ASE** will comprise of all the requirements documented in the current document and will be operated in the manner specified in the document. The Source code will be developed subsequently based on these requirements and will formally go through code review during testing process.

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Software Requirement Specification (SRS)

❖ Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to identify and document the user requirements for Quantum-Proof Systems Scanner. This document provides a comprehensive description of the system requirements, expected functionality, and operational behavior of the proposed solution. It acts as a formal reference for stakeholders including system developers, cybersecurity teams, administrators, and management involved in the planning, design, development, and deployment of the system.

The SRS establishes a clear understanding of how the system will operate, the services it will provide, and the constraints under which it will function. It also serves as a baseline for system design, implementation, testing, and future enhancements.

This document is prepared with the following objectives:

- To describe the behavior and operational functionality of the system.
- To define the functional and non-functional requirements of the system.
- To provide process flow charts and workflow representations for understanding system operations.
- To provide a structured reference for system design, development, testing, and deployment.
- To ensure that the system meets the required cybersecurity, compliance, and operational standards for identifying and managing cryptographic configurations in public-facing applications.

1.2 Scope

The proposed system Quantum-Proof Systems Scanner is designed to enhance the cybersecurity posture of public-facing banking applications by identifying cryptographic implementations and evaluating their readiness against emerging quantum computing threats. The system will function as an automated scanning and monitoring platform capable of discovering cryptographic assets, analyzing security configurations, and providing actionable insights for improving cryptographic resilience.

The scope of the system includes the following functionalities:

- Discover cryptographic inventory associated with public-facing applications, including TLS certificates, VPN endpoints, and APIs exposed to the internet.
- Identify and analyze cryptographic controls implemented within the system, such as cipher suites, key exchange mechanisms, encryption algorithms, and TLS protocol versions.
- Validate whether the deployed cryptographic algorithms and configurations comply with Post-Quantum Cryptography (PQC) standards recommended by the National Institute of Standards and Technology (NIST).
- Generate detailed Cryptographic Bill of Materials (CBOM) inventory for identified assets in accordance with guidelines provided by Indian Computer Emergency Response Team (CERT-In).
- Provide actionable recommendations and remediation guidance for assets that are not compliant with PQC standards or contain weak cryptographic configurations.
- Issue automated digital labels and security status indicators such as Quantum-Safe, PQC Ready, or Fully Quantum Safe based on the cryptographic posture of each scanned asset.
- Provide an enterprise-wide centralized management console through a secure graphical user interface (GUI) that allows administrators and cybersecurity teams to monitor, manage, and analyze the cryptographic status of all scanned public-facing applications.
- Display risk ratings and security scores (High, Medium, Low) for each application based on its cryptographic configuration and compliance status. The dashboard dynamically reflects changes in security posture whenever cryptographic configurations are updated or vulnerabilities are identified.
- Enable reporting and visualization of scanning results, cryptographic inventory, compliance status, and recommendations through the centralized dashboard interface.

1.3 Intended Audience

The intended audience of this document includes business stakeholders and technical users from Punjab National Bank (PNB) who are involved in cybersecurity operations, system administration, compliance management, and digital infrastructure management.

This document is primarily intended for the following groups:

- Cybersecurity Teams responsible for monitoring and improving the security posture of banking applications.

- System Administrators and IT Operations Teams responsible for deploying, configuring, and maintaining the system infrastructure.
- Compliance and Risk Management Teams responsible for ensuring that banking systems comply with security standards and regulatory requirements.
- Software Developers and System Architects involved in the design and development of the proposed solution.
- Project Managers and Decision Makers who require an overview of the system's capabilities, scope, and operational objectives.

The document serves as a reference guide for understanding the system requirements, operational workflow, and security capabilities of the proposed solution.

❖ Overall Description

2.1 Product Perspective

The proposed system **Quantum-Proof Systems Scanner** is designed as an advanced cybersecurity solution aimed at identifying, analyzing, and managing cryptographic implementations used in public-facing banking applications. It functions as a centralized scanning and monitoring platform capable of detecting cryptographic configurations, evaluating their security posture, and assessing their readiness against emerging threats posed by quantum computing.

The system operates as a **security assessment and compliance tool** that integrates with existing banking infrastructure without disrupting live services. It continuously analyzes **internet-exposed assets**, including:

- Web servers
- APIs
- VPN gateways
- Other externally accessible services

Through this analysis, the platform detects and evaluates:

- Cryptographic protocols
- SSL/TLS certificate configurations
- Encryption mechanisms used in secure communication

The proposed solution follows a **modular architecture** consisting of several specialized components:

- **Asset Discovery Module** – identifies and catalogs internet-facing assets.
- **Cryptographic Scanner** – detects cryptographic protocols, cipher suites, and certificates in use.

- **Post-Quantum Cryptography (PQC) Validator** – evaluates the readiness of current cryptographic systems against quantum computing threats.
- **Cryptographic Bill of Materials (CBOM) Generator** – generates a structured inventory of cryptographic components used across applications.
- **Enterprise Monitoring Dashboard** – provides a centralized interface for monitoring risks, viewing analytics, and generating compliance reports.

The system aligns with **Post-Quantum Cryptography (PQC) recommendations from the National Institute of Standards and Technology (NIST)** and generates cryptographic inventories following **CBOM guidelines issued by the Indian Computer Emergency Response Team (CERT-In)**.

By enabling **automated detection, classification, and reporting of cryptographic assets**, the system helps organizations proactively prepare for the transition to **quantum-safe cryptographic systems**. The product ultimately functions as an **enterprise-level monitoring platform** that provides centralized visibility, risk evaluation, and compliance reporting for public-facing applications within the banking ecosystem.

2.2 Product Functions

The proposed system will perform a set of advanced functions designed to detect cryptographic implementations and evaluate their security posture in relation to quantum-safe standards. The major functions of the system include the following:

- **Asset Discovery**: Identify and catalogue public-facing applications such as web servers, APIs, VPN gateways, and other internet-exposed services.

- **Cryptographic Inventory Discovery**: Extract and document cryptographic elements including TLS certificates, encryption algorithms, cipher suites, key exchange mechanisms, and TLS protocol versions used by the identified assets.

- **Cryptographic Analysis**: Analyze the detected cryptographic configurations to identify weak, deprecated, or non-compliant encryption mechanisms.

- **Post-Quantum Cryptography Validation**: Evaluate cryptographic algorithms against PQC standards defined by the National Institute of Standards and Technology (NIST) to determine their resistance against potential quantum computing attacks.

- **CBOM Generation**: Automatically generate a Cryptographic Bill of Materials (CBOM) containing a detailed inventory of cryptographic assets and configurations as recommended by Indian Compute Emergency Response Team (CERT-In).

- **Risk Assessment and Scoring**: Assign risk ratings such as High, Medium, or Low to each application based on the strength of its cryptographic configuration and its compliance with PQC standards.

- **Recommendation Engine**: Provide remediation guidance and actionable recommendations for systems that are not compliant with quantum-safe cryptographic standards.

- **Quantum-Safe Certification**: Automatically issue digital security labels such as Quantum-Safe, PQC Ready, or Fully Quantum Safe for applications that meet predefined cryptographic requirements.

- **Enterprise Monitoring Dashboard**: Provide a centralized graphical console that displays the real-time security posture, scanning results, cryptographic inventory, and compliance status of all monitored systems.

- **Reporting and Exporting**: Generate detailed security reports and export them in machine-readable formats such as JSON, XML, and CSV for integration with other security monitoring systems.

2.3 User Classes and Characteristics

The proposed system **Quantum-Proof Systems Scanner** will be accessed by different categories of users within the organization, each having specific responsibilities and access privileges. The users are expected to possess varying levels of technical knowledge related to networking, cybersecurity, and cryptographic systems. Access to system features will be controlled through role-based access mechanisms to ensure secure and controlled usage of the platform.

The major user classes and their characteristics are described below:

Primary Users

- **Bank Cybersecurity Teams:**

  These users are responsible for monitoring the security posture of public-facing banking applications. They use the system to perform cryptographic scans,

analyze TLS configurations, detect vulnerabilities, and evaluate the readiness of systems against quantum computing threats. They are expected to have strong technical knowledge of cybersecurity practices, cryptographic protocols, and network security mechanisms.

- **IT Administrators:**

  IT administrators manage system configuration, infrastructure deployment, and overall operational management of the platform. They are responsible for configuring scanning parameters, maintaining system availability, managing user access privileges, and ensuring the proper functioning of the system within the banking network environment.

Secondary Users

- **Compliance Auditors:**

  Compliance auditors use the system to review security reports, CBOM inventories, and compliance status of public-facing applications. Their primary role is to ensure that cryptographic implementations follow security standards and regulatory guidelines. They generally require read-only access to reports and dashboards.

- **Risk Managers:**

  Risk management teams analyze security risk levels associated with cryptographic configurations identified by the system. They evaluate risk ratings such as High, Medium, or Low and use the generated recommendations to plan mitigation strategies and improve the organization's overall cybersecurity posture.

User Type	Organization	Description	Menus / Access Rights
Admin User	PNB / IIT Kanpur Officials	Responsible for system configuration, user management, scan scheduling, and overall monitoring of the platform.	Full system access including configuration, scanning, reporting, and dashboard management.
PNB Checker	PNB Cybersecurity Team	Responsible for performing cryptographic scans, analyzing results, and reviewing CBOM reports for public-facing applications.	Access to scanning tools, asset inventory, risk reports, and monitoring dashboard.

2.4 Operating Environment

The operating environment for the **Quantum-Proof Systems Scanner** consists of the hardware and software infrastructure required for the deployment, execution, and operation of the system. The system will be deployed within a secure banking network environment and will support centralized monitoring and scanning of public-facing applications.

The operating environment for the system is listed below:

- **Server System**: High-availability enterprise server or virtual machine infrastructure capable of supporting large-scale network scanning and cryptographic analysis. The server must support secure communication and high processing capability to handle multiple scanning tasks simultaneously.
- **Operating System**: Linux-based operating systems such as Ubuntu Server, Red Hat Enterprise Linux (RHEL), or CentOS are recommended due to their stability, security features, and compatibility with cybersecurity tools.
- **Database**: A relational database management system such as PostgreSQL or MySQL will be used to store cryptographic inventory data, scan results, CBOM records, user activity logs, and security reports.
- **Platform**: The system will operate as a web-based application platform accessible through a secure browser interface. It will support centralized access for administrators and cybersecurity teams through an enterprise dashboard.

- **Technology**: The system will be developed using modern web technologies and cybersecurity tools such as Python / Java for backend processing, RESTful services for communication, and HTML5, CSS, and JavaScript for the frontend interface. Cryptographic analysis tools and security libraries will be integrated to support TLS scanning and algorithm validation.

- **API**: The system will utilize secure REST APIs for communication between modules, integration with external security tools, and interaction with the enterprise dashboard. APIs will also enable automated data exchange with other security monitoring systems if required.

- This operating environment ensures that the proposed system can operate efficiently, securely, and reliably within the banking infrastructure while supporting advanced cryptographic scanning and monitoring capabilities.

2.5 Design and Implementation Constraints

The design and implementation of the proposed system **Quantum-Proof Systems Scanner** must follow certain technical, security, performance, and usability constraints to ensure that the solution operates effectively within the banking infrastructure while maintaining high standards of reliability, security, and compliance.

1. Technical Constraints (For Deployment)

- **Network Configuration:**
  The application must support private (intranet) IP addressing and operate within the secure banking network environment. Appropriate firewall configurations, routing rules, and network security controls must be implemented to allow controlled scanning of public-facing assets without exposing internal systems.

- **Hosting Environment:**
  The system should be deployed within the bank's internal intranet infrastructure or a secure enterprise server environment. External access to the system should be restricted and controlled through secure authentication mechanisms.

- **Integration Constraints:**
  The system must be capable of integrating with existing banking infrastructure such as security monitoring systems, asset management tools, and compliance reporting systems without disrupting existing operations.

2. **Security Constraints**

- **Access Control:**

  The system must implement Role-Based Access Control (RBAC) to ensure that users can only access functionalities relevant to their roles, such as administrator access, security analyst access, or auditor access.

- **Data Encryption:**

  All data transmitted between system components and users must be protected using secure communication protocols such as HTTPS with TLS encryption to ensure confidentiality and integrity of sensitive information.

- **Authentication and Authorization:**

  Secure authentication mechanisms must be implemented to verify user identity and ensure authorized access to system resources.

2.6 Assumptions and Dependencies

- **Assumptions:**

  Standard Browser Support- e.g. It is assumed that end users will be accessing the application using HTML5-compliant browser such as Google Chrome.

- **Dependencies:**

  Database System-e.g. The application is dependent on Oracle database for data storage. Any maintenance, downtime, or performance issues with the database will directly impact on the application's functionality.

  Examples:

  - Assumes TLS-based communication is used in all public-facing applications.
  - Depends on NIST PQC algorithms being standardized and available.
  - Assumes internet connectivity for scanning endpoints.

❖ <u>Specific Requirements</u>

3.1 Functional Requirements

The functional requirements describe the core operations and services that the system **Quantum-Proof Systems Scanner** must perform. These requirements define how the system will detect cryptographic configurations, analyze security posture, and provide actionable insights regarding quantum-safe readiness of public-facing banking applications.

The major functional requirements of the system are as follows:

- **Asset Discovery**: The system shall identify and catalogue public-facing applications including web servers, APIs, and VPN endpoints that are exposed to the internet.

- **Cryptographic Inventory Discovery**: The system shall automatically detect cryptographic elements such as TLS certificates, encryption algorithms, cipher suites, key exchange mechanisms, and TLS protocol versions used by the identified assets.

- **Cryptographic Configuration Analysis**: The system shall analyze detected cryptographic configurations to identify weak, deprecated, or insecure encryption mechanisms.

- **Post-Quantum Cryptography Validation**: The system shall evaluate the detected cryptographic algorithms against Post-Quantum Cryptography standards recommended by the National Institute of Standards and Technology (NIST) to determine whether the asset is quantum-safe.

- **CBOM Generation**: The system shall generate a Cryptographic Bill of Materials (CBOM) inventory that lists all cryptographic components and configurations associated with public-facing applications in accordance with guidelines provided by Indian Computer Emergency Response Team (CERT-In).

- **Risk Assessment and Rating**: The system shall assign risk levels such as High, Medium, or Low to each application based on the strength of its cryptographic configuration and compliance with PQC standards.

- **Recommendation Engine**: The system shall provide actionable recommendations and remediation guidance for systems that are not compliant with quantum-safe cryptographic standards.

- **Dashboard Monitoring**: The system should provide a centralized dashboard that displays cryptographic inventory, security status, risk ratings, and compliance information for all scanned assets.

- **Report Generation**: The system shall generate detailed reports containing cryptographic inventory, vulnerability analysis, compliance status, and security recommendations.

- **Security Label Generation**: The system shall automatically issue digital labels such as Quantum-Safe, PQC Ready, or Fully Quantum Safe for applications that meet predefined security criteria.

3.2 External Interface Requirements

3.2.1 User Interfaces

The application shall provide a web-based user interface accessible via modern web browsers such as Google Chrome, Microsoft Edge, or Mozilla Firefox.

The interface shall include the following components:

Login Interface: Secure authentication interface for system access.

Dashboard Interface: Displays system overview including scanned assets, cryptographic inventory, and security status.

Scan Configuration Interface: Allows administrators or analysts to initiate or schedule scans for public-facing applications.

Reports Interface: Displays detailed reports including CBOM inventory, risk scores, and recommendations.

Security Status Indicators: Visual representation of security posture using risk ratings such as High, Medium, or Low.

The user interface shall be designed to ensure ease of navigation, clarity of information, and efficient interaction with system functionalities

3.3 System Features

- Automated Asset Discovery to identify public-facing applications and internet- exposed services.
- Cryptographic Scanning Engine to detect encryption algorithms, certificates, and cipher suites.
- Post-Quantum Cryptography Validator to evaluate algorithms against PQC standards.
- Cryptographic Bill of Materials (CBOM) Generator to create detailed inventories of cryptographic assets.
- Risk Assessment Engine to calculate and assign risk ratings based on cryptographic strength.
- Recommendation Engine to provide guidance for replacing insecure algorithms with quantum-safe alternatives.
- Enterprise Monitoring Dashboard to visualize cryptographic status and security posture of all assets.
- Automated Reporting System to generate security reports for compliance and audit purposes.

3.4 Non-functional Requirements

Non-functional requirements define system characteristics such as performance, reliability, security, and usability.

3.4.1 Performance Requirements

- The system shall support scanning of multiple public-facing assets simultaneously without significant performance degradation.
- Cryptographic scanning and analysis should be completed within an acceptable time frame depending on the number of assets being analysed.
- The dashboard should update scan results and security status in near real-time.
- The system should support scalable architecture to accommodate increasing numbers of assets and users.

3.4.2 Software Quality Attributes

- Security: Protection of sensitive data through secure communication protocols and strong authentication mechanisms.
- Reliability: Ability to operate consistently without failures during scanning and analysis processes.
- Scalability: Capability to support growing infrastructure and increasing numbers of monitored assets.
- Maintainability: Modular architecture enabling easy updates and enhancements.
- Usability: User-friendly interface enabling cybersecurity teams to easily monitor system security posture.

3.4.3 Other Non-functional Requirements

- The system shall maintain detailed logs of all scanning activities and user actions for auditing purposes.
- The system shall support generation of reports in machine-readable formats such as JSON, XML, and CSV.
- The system shall support secure backup and recovery mechanisms to protect stored data.
- The system shall maintain high availability and support disaster recovery procedures to ensure continuous system operation.

4. Technological Requirements

The proposed system **Quantum-Proof Systems Scanner** will be developed using modern web technologies and cybersecurity tools to ensure reliability, scalability, and secure operation within the banking environment. The system will follow a modular architecture consisting of frontend, backend, scanning engine, and database components.

4.1 Technologies used in development of the web application

- **Backend Development:**

Python / Java will be used for implementing the core application logic, cryptographic analysis modules, and scanning engine responsible for discovering and analyzing public-facing assets.

- **Frontend Development:**

HTML5, CSS3, and JavaScript will be used to design an interactive and responsive web-based user interface for the dashboard and management console.

- **Web Framework:**

Frameworks such as Django / Flask (Python) or Spring Boot (Java) may be used to develop secure and scalable web services.

- **API Communication:**

RESTful APIs will be used for communication between system modules and integration with external systems or security monitoring platforms.

- **Cryptographic and Security Libraries:**
  OpenSSL libraries and TLS analysis tools will be integrated to detect certificate information, encryption algorithms, cipher suites, and protocol versions used by public-facing applications.

4.2 I.D.E. (Integrated Development Environment):

The development of the system will be carried out using modern Integrated Development Environments (IDEs) that support efficient coding, debugging, testing, and deployment.

Examples of IDEs that may be used include:

- Visual Studio Code – for frontend and backend development
- IntelliJ IDEA – for Java-based backend development
- Eclipse IDE – for enterprise-level application development
- PyCharm – for Python-based application development

These development environments provide support for debugging tools, version control integration, and project management features required during the development lifecycle.

4.3 Database Management Software:

The system will use a Relational Database Management System (RDBMS) to store cryptographic inventory data, scanning results, user information, system logs, and compliance reports.

Possible database systems include:

- PostgreSQL – for reliable and scalable enterprise database management
- MySQL – for structured data storage and efficient query processing
- Oracle SQL – for enterprise-grade database deployment in large banking infrastructures

The database will maintain records related to asset inventory, cryptographic configurations, CBOM data, risk ratings, and historical scan reports. Proper database security measures such as access control, encryption, and backup mechanisms will be implemented to ensure data integrity and confidentiality

5. Security Requirements

Security is a critical component of the proposed system <Project Name>, as it operates within a sensitive banking environment and interacts with public-facing applications. The system must ensure protection of data, secure communication, controlled access, and compliance with organizational and regulatory cybersecurity standards.

The following security requirements shall be considered during the design, development, deployment, and operation of the system.

Compatibility with Existing IT Infrastructure

The proposed system must be compatible with the existing IT infrastructure of the organization. Integration with current systems, networks, and security tools should be performed without disrupting ongoing banking operations. The implementation should ensure that existing systems remain unaffected and continue functioning normally during and after the deployment of the proposed solution.

Audit Trails and Logging

The system should maintain comprehensive audit trails for all critical operations and user activities. Logs must capture important details including:

- User ID
- Date and time of activity
- Type of operation performed
- System response and status

All scanning results, API responses, system events, and user actions shall be recorded and stored securely in the database for monitoring, analysis, and auditing purposes.

Access Control and Authorization

The system shall implement strict Role-Based Access Control (RBAC) to ensure that access to system functionalities and sensitive information is restricted based on user roles and responsibilities.

Security principles such as segregation of duties and need-to-know access must be enforced. For example:

- Only Admin Users will have permission to configure system settings and schedule scans.

- Security Analysts / PNB Checkers will be able to initiate scans and view reports.
- Compliance Auditors will have read-only access to reports and dashboards.

This ensures that unauthorized users cannot access sensitive data or modify system configurations.

Recoverability and Disaster Recovery

The system must support mechanisms for data recovery and disaster recovery (DR) in case of system failure, data corruption, or security incidents. Regular backups of system data, logs, and configuration settings must be maintained to ensure that the system can be restored to a functional state with minimal data loss.

❖ **Compliance with Legal and Regulatory Requirements**

The system must comply with applicable legal, statutory, and contractual obligations related to cybersecurity, data protection, and banking regulations. The system must follow industry security standards and regulatory frameworks applicable to financial institutions.

❖ **Security Risks when Connecting with External Systems**

When interacting with external systems or public-facing applications, the system must ensure that communication channels are secure and protected against potential vulnerabilities. Any security vulnerabilities related to integration with external systems shall be identified through regular security assessments and audits.

❖ **Operating Environment Security**

The operating environment of the system must maintain strong security controls including secure communication protocols. All data communication between system components and users must be protected using HTTPS with TLS encryption (TLS 1.2 or higher) to ensure confidentiality and integrity of transmitted information.

❖ **Cost of Security Implementation**

The overall cost of implementing and maintaining security measures throughout the lifecycle of the system must be considered. This includes costs related to:

- Hardware infrastructure required for secure deployment
- Software tools and security frameworks
- Personnel responsible for system monitoring and maintenance
- Training programs for users and administrators and resource

Annexure-A (CERT-In CBOM Elements)

The system shall generate a Cryptographic Bill of Materials (CBOM) containing minimum cryptographic asset elements as recommended by Indian Computer Emergency Response Team (CERT-In). The CBOM helps identify cryptographic algorithms, keys, protocols, and certificates used by public-facing applications.

The minimum CBOM elements are described below.

1. **Algorithms**

Cryptographic Asset Type	Element	Description
Algorithms	Name	Name of the cryptographic algorithm or asset. Example: AES-128-GCM.
Algorithms	Asset Type	Specifies type of cryptographic asset
Algorithms	Primitive	Describes cryptographic primitive such as SHA512withRSA.
Algorithms	Mode	Operational mode used by the algorithm such as GCM for AES encryption.

2. **Cryptographic Functions**

Cryptographic Asset Type	Element	Description
Crypto Functions	Crypto Functions	Lists functions supported such as key generation, encryption, decryption, and authentication tag generation.
Crypto Functions	Classical Security Level	Indicates the classical security strength of the algorithm (e.g., AES-128 = 128 bits).
Crypto Functions	OID	Object Identifier used to uniquely identify cryptographic algorithms.
Crypto Functions	List	List of cryptographic algorithms supported by the system.

3. Keys

Cryptographic Asset Type	Element	Description
Keys	Name	Name of the cryptographic key.
Keys	Asset Type	Type of asset (key).
Keys	ID	Unique identifier or reference number for the
Keys	State	State of the key such as active, revoked, or
Keys	Size	Size of the key measured in bits (e.g., 128-bit AES key, 2048-bit RSA key).
Keys	Creation Date	Date when the key was created.
Keys	Activation Date	Date when the key became operational.

4. Protocols

Cryptographic Asset Type	Element	Description
Protocols	Name	Name of the cryptographic protocol such as TLS, IPSec, or SSH.
Protocols	Asset Type	Specifies the asset type as protocol.
Protocols	Version	Version of the protocol used such as TLS 1.2 or TLS 1.3.
Protocols	Cipher Suites	Set of cryptographic algorithms supported by the protocol.
Protocols	OID	Object Identifier associated with the protocol specification.

5. Certificates

Cryptographic Asset Type	Element	Description
Certificates	Name	Name of the certificate representing the entity (e.g., website).
Certificates	Asset Type	Specifies the asset type as certificate.
Certificates	Subject Name	Distinguished Name (DN) of the entity identified by the certificate.
Certificates	Issuer Name	Certificate Authority that issued the certificate.
Certificates	Not Valid Before	Date from which certificate validity
Certificates	Not Valid After	Expiry date of the certificate.
Certificates	Signature Algorithm Reference	Algorithm used to sign the certificate with its OID.
Certificates	Subject Public Key Reference	Reference to the subject public key and its algorithm.
Certificates	Certificate Format	Format of certificate such as X.509.
Certificates	Certificate Extension	File extension of certificate such as .crt.